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Precision through progression: Empowering temporal knowledge graph reasoning with knowledge-guided chain of thought

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ABSTRACT

Temporal Knowledge Graphs (TKGs) have emerged as a powerful paradigm for event forecasting, owing to their ability to dynamically represent the evolving relationships between entities over time. By effectively reasoning along the temporal dimension, TKGs help address real-world data incompleteness through inference of missing facts. Recent advances in large language models (LLMs) have led to their integration with TKG reasoning tasks. However, current LLM-based approaches face three critical challenges: (1) insufficient utilization of background knowledge, (2) inadequate modeling of the evolving temporal dynamics intrinsic to TKGs, and (3) difficulty in bridging the structural mismatch between the graph structure and the sequential operation mode of LLMs. To address these challenges, we propose EV-COT, a novel EVent-aware Chain-Of-Thought reasoning framework designed to explicitly model event evolution through structured, interpretable reasoning chains. EV-COT comprises three modular, plug-and-play components – knowledge module, perception module, and thinking module – that work collaboratively to extract essential event-related cues for enhanced reasoning. Specifically, the knowledge module generates high-quality contextual knowledge to enrich entity representation, and the perception module captures intricate structural and temporal patterns inherent in TKGs. Moreover, the thinking module extracts temporal logical rules, facilitating interpretable step-by-step reasoning. By effectively integrating these diverse contextual knowledge, EV-COT delivers more accurate predictions. Extensive evaluations on three datasets demonstrate that EV-COT consistently outperforms state-of-the-art methods, highlighting its effectiveness for precise event forecasting in TKGs.

1. Introduction

The rapid evolution of internet and mobile communication technologies has led to an explosion of textual data, capturing the complexities of real-world events and their interconnections. Knowledge Graphs (KGs) [1], structured representation of factual events using *(subject, relation, object)* triplets, have enabled a plethora of knowledge-driven applications in diverse domains, including natural language understanding [2], recommender systems [3,4], and social network analysis [5–7]. However, the static nature of traditional KGs limits their ability to accurately reflect the dynamic nature of real-world events. Consider a factual event: *(François Hollande, president Of, France)*, it is now outdated because *Emmanuel Macron* has been the President of the France since 2021. To address this temporal limitation, Temporal Knowledge Graphs (TKGs) have been introduced. TKGs incorporate temporal information by representing facts as quadruples: *(subject, relation, object, time)*. Consequently, the aforementioned example becomes *(François Hollande, presidentOf, France, 2012/05/15-2017/05/14)*, accurately reflecting the time period during which the statement held true. Essentially, a TKG comprises multiple KG snapshots, each representing a specific timeframe within which factual events co-occur.

The inherent incompleteness of TKGs necessitates reasoning mechanisms to infer missing factual events at specific points in time [8–10]. This task can be broadly categorized into two settings: *interpolation* and *extrapolation* [11,12]. In the interpolation setting [13], the goal is to infer the missing factual information at a specific time t, given a TKG with timestamps ranging from t_0 to t_T, where $t_0 < t < t_T$. In contrast, extrapolation [11,14] tackles the more challenging task of reasoning about facts occurring at future timestamps beyond the observed range t_T. Extrapolation task is inherently more difficult as it involves predicting events

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without the benefit of complete historical contexts. Our research focuses primarily on addressing the challenges of extrapolation setting in TKG reasoning, aiming to predict unknown future facts.

To address the challenges in Temporal Knowledge Graph Reasoning (TKGR), existing supervised methods [9,11,15,16] predominantly employ graph neural networks (GNNs) [17,18] to model interactive relationships among factual events and derive entity and relation representations. Subsequently, recurrent neural networks (RNNs) [19] are commonly used to capture temporal dependencies, learning how temporal information propagates and influences future states. With the advent of Large Language Models (LLMs), researchers have begun exploring ways to leverage their extensive world knowledge and robust semantic reasoning capabilities to enhance performance on TKGR task. LLM-based methods [10,20,21] for TKGR primarily prompt LLMs with historical events, leveraging their reasoning abilities to infer new facts and deliver strong reasoning performance. Consequently, these approaches have garnered considerable attention in the research community.

Despite their success, these methods still face several challenges: **(1) Lack of background knowledge**. In TKGs, entities are typically *textless* nodes that lack explicit semantic information. Existing methods primarily captures structural and temporal dependencies in TKGs while disregarding the entity background information, which weakens the correlations between facts during reasoning [22]. Therefore, these models often struggle to capture nuanced semantic correlations between events, potentially leading to inaccurate predictions. **(2) Evolving Properties in TKGs**. In reality, the knowledge stored in TKGs continuously evolves over time, resulting in temporal distribution shifts between initial observations and future events [23]. However, continuously updating LLM-based methods to incorporate evolving knowledge from TKGs is impractical due to the substantial computational resources required for fine-tuning [24]. **(3) Gap Between Graphs and LLMs**. Relying solely on the reasoning capabilities of LLMs remains insufficient for the TKGR task. While LLMs demonstrate strong semantic reasoning capabilities, they inherently struggle to capture complex structural dependencies and temporal interactions within TKGs. This limitation results in the loss of critical evidence and negatively impacts reasoning performance.

To address these challenges, we draw inspiration from the remarkable human ability to navigate new scenarios [25], which involves understanding context, acquiring relevant background knowledge, and integrating these insights for logical reasoning and planning. Motivated by these observations, we propose **EV-COT**, a novel **EV**ent-aware **C**hain-**O**f-**T**hought reasoning framework that constructs step-by-step reasoning chains to enhance event evolution modeling. Specifically, the event-aware COT involves retrieving background knowledge, modeling structural-temporal dependencies, and adaptively capturing temporal logical rules in a hierarchical reasoning manner. Our EV-COT comprises three key modules: a **knowledge** module, a **perception** module, and a **thinking** module.

In the knowledge module, we devise an entity-oriented knowledge *retrieval* and *generation* strategy to obtain structured commonsense knowledge for each entity. Furthermore, we craft a knowledge prompt that enables LLMs to effectively filter and refine this commonsense, producing in high-quality background knowledge. In the perception module, we primarily capture crucial structural dependencies and temporal patterns from the perspectives of knowledge graph topology and sequential snapshots. We first employ a pre-trained language model to encode background knowledge and then utilize a relation-aware graph encoder-enriched with this knowledge to extract key event evidence across historical KG snapshots. Subsequently, in the thinking module, EV-COT focuses on analyzing historical data and mining temporal logical rules, unveiling temporal patterns and enabling interpretable reasoning. We introduce an adaptive update strategy that iteratively updates the temporal rules extracted by LLMs. These rules are continuously refined and ranked to incorporate the latest knowledge, thereby improving predictions for future events. Finally, we combine the outputs from the perception and thinking modules to generate a more comprehensive and accurate final prediction. Overall, the main contributions are summarized as follows:

- We are the first to emphasize and systematically investigate the necessity of capturing background knowledge, evolving properties, and the inherent gap between graph structures and language space in TKGR. To this end, we propose EV-COT, a novel event-aware chain-of-thought reasoning framework designed to tackle these challenges and enhance TKGR through step-by-step reasoning.
- We propose three plug-and-play reasoning stages- *knowledge*, *perception*, and *thinking* - that collaboratively capture comprehensive event correlation cues, encompassing contextual background knowledge, structural-temporal dependencies, and temporal logical rules. This hierarchical reasoning further integrates graph-centric and LLM-centric knowledge, thereby enabling accurate and interpretable reasoning without requiring additional training data.
- We conduct extensive experiments on three publicly available benchmark datasets. The results demonstrate that EV-COT consistently outperforms state-of-the-art models on the TKGR task, thereby validating the effectiveness of our design for event-aware chain-of-thought reasoning.

2. Related work

2.1. Temporal knowledge graph reasoning

Recently, TKGR has received considerable research attention, with existing approaches generally categorized into two settings: interpolation and extrapolation [22,26,27].

2.1.1. Interpolation

Interpolation reasoning aims to exploit missing historical facts existing in the TKG. Most exsisting works, such as HyTE [28], TNTComplEx [29], TTransE [30] and ChronoR [31], have proposed diverse and effective frameworks aimed at addressing key challenges in the interpolation setting of TKGR. Specifically, TTransE [30] considers the time information, and meanwhile treats both relation and time as translations between entities. ChronoR [31] learns a k-dimensional rotation transformation parametrized by relation and time to capture temporal interactive information. However, these models are not well-suited for the extrapolation setting, as they inherently struggle to model evolving structural dependencies and tackle temporal distribution shifts when predicting future events.

2.1.2. Extrapolation

The task of TKGR under the extrapolation setting aims to infer future events according to historically associated facts, which is more challenging and pragmatic [32–35]. CyGNet [36] introduces the copy mechanism to concentrate on events of reoccurrence. CENET [9] designs a contrastive learning framework to capture historical and non-historical dependency. In addition, other works mainly combine GNNs and RNNs to jointly model event structural information and temporal interactive dependencies. For example, RE-NET [15] utilizes a neighborhood aggregator to model connecting relations of concurrent facts and a recurrent encoder to capture event sequential patterns. RE-GCN [11] captures the evolutional representations of entities and relations by RGCN [37] and employs the Gate Recurrent Unit (GRU) to learn event sequential patterns. TiRGN [8] models the intricate sequential, repetitive, and cyclical patterns of events through a time-guide recurrent graph network. xERTE [38] uses a subgraph sampling technique to build an inference graph. Tlogic [39] utilizes the temporal random walks and extracts the temporal logical rules, leading to better explainability. TANGO [40] extends the concept of ordinary differential equations (ODEs) to multi-graph convolutional networks to model TKGs. RETIA [41] constructs twin hyper-relation subgraphs to enhance the representation learning of relations and entities. L^2TKG [16] extracts intra-time latent relations among co-occurring entities and inter-time latent relations between entities. TiPNN [42] learns historical factual information from an entity-independent perspective. IE-Evo [43] captures both internal and external evolution-enhanced information to improve performance on TKGR.

However, existing methods primarily focus on modeling two types of information, namely intra-snapshot structural information and inter-snapshot temporal interactions. Typically, they overlook valuable prior background knowledge associated with entities as well as temporal logical rules. To address this limitation, we propose an event-aware chain-of-thought reasoning framework that collaboratively captures a comprehensive spectrum of event correlation cues-including contextual background knowledge, structural-temporal dependencies, and temporal logical rules-by incorporating step-by-step reasoning chains.

2.2. Large language model for TKGR

Pre-trained language models (PLMs), such as BERT [44] and GPT [45], are first trained on large-scale unlabeled text corpora and then fine-tuned to adapt to specific tasks. Due to their ability to capture contextual semantic information, PLM-based models have been increasingly explored for temporal knowledge graph reasoning in recent years. KG-BERT [46] directly combines the subject, predicate, and object from triples to form a text-based input for the PLM. PPT [47] converts the TKGR task as a masked token prediction problem using a PLM and designs specific prompts for different types of intervals between timestamps to capture fine-grained semantic information from temporal data.

Recently, *large language models* (LLMs) [48] have exhibited advanced language understanding and generation capabilities across a wide range of text-related tasks [49]. LLMs are typically pre-trained in an autoregressive manner using a next-word prediction objective [50] and exhibit strong proficiency in text comprehension and generation. Thus, in the field of TKGs, integrating LLMs into the task of TKGR has become a prominent trend. zrLLM [20] converts the graph structure into a textual format and inputs these text-encoded relations into LLMs to generate corresponding descriptions. These descriptions are then incorporated into embedding-based models to supplement the semantic information of zero-shot relations. GPT-NeoX [10] reformulates TKG prediction as an in-context learning (ICL) task, providing LLMs with the first-order history of the query in textual form to generate possible answers without any fine-tuning. Mixtral-CoH [21] employs a chain-of-history reasoning mechanism to iteratively extract high-order historical information.

However, existing LLM-based models often overlook valuable prior knowledge and the intrinsic evolving structural-temporal patterns inherent in TKGs. To address this gap between graph structures and the language space, we propose an event-aware chain-of-thought reasoning framework that further integrates graph-centric and LLM-centric knowledge, including entity-oriented information, topology-aware structural dependencies, temporal patterns, and temporal logical rules, to enhance event evolution modeling.

3. Preliminaries

In this section, we provide the necessary background and formulate temporal knowledge graph reasoning (TKGR) task.

Temporal Knowledge Graph. Let $\mathcal{E}$ and $\mathcal{R}$ represent the sets of entities and relations, respectively. A quadruple (s, r, o, t) represents that a subject entity $s \in \mathcal{E}$ has a relation $r \in \mathcal{R}$ with an object entity $o \in \mathcal{E}$ at a specific timestamp t. All quadruples (a.k.a. facts) occurring within the time interval $(t - \Delta t, t]$, which is denoted simply as t for brevity, collectively form a knowledge graph G_t. A temporal knowledge graph, denoted as $\mathcal{G} = \{G_0, G_1, \ldots, G_{t-1}\}$, consists of a sequence of knowledge graphs arranged in chronological order.

Temporal Knowledge Graph Reasoning. The TKGR primarily falls into two settings: interpolation and extrapolation. Interpolation focuses on inferring incomplete or missing historical facts, whereas extrapolation aims to predict future facts based on existing TKG knowledge. Under the extrapolation setting, TKGR typically involves two sub-tasks: entity prediction and relation prediction. In the context of extrapolation, entity prediction targets the prediction of a missing object entity in a query $(s, r, ?, t)$ or a missing subject entity in a query $(?, r, o, t)$, given the historical TKG $\mathcal{G}$. Similarly, relation prediction seeks to infer the missing relation in a query $(s, ?, o, t)$.

In this work, we mainly focus on the task of entity prediction under the extrapolation setting. Furthermore, we have observed that existing works predominantly leverage two kinds of information (i.e., intra-snapshot structural information and inter-snapshot temporal interactions) to construct evolutional embeddings of entities $\mathbf{H}_t \in \mathbb{R}^{|\mathcal{E}|\times d}$ and the relation $\mathbf{R}_t \in \mathbb{R}^{|\mathcal{R}|\times d}$ at the timestamp t. Formally, given a query $(s, r, ?, t)$ and historical TKG $\mathcal{G}$, the task of TKGR aims to predict the conditional probability of all object entities at the timestamp t with the evolutional entity representations $\mathbf{H}_t$ and the relation representations $\mathbf{R}_t$, which can be summarized as follows:

$$f(o \mid s, r, t, \mathcal{G}) = f\left(o \mid s, r, \mathbf{H}_t, \mathbf{R}_t\right). \tag{1}$$

4. Methodology

As depicted in the Fig. 1, EV-COT is composed of three key components: a knowledge module, a perception module, and a thinking module. In the following parts, we will provide a detailed description to each component of our EV-COT.

4.1. Knowledge - background knowledge construction

Background knowledge typically offers valuable auxiliary information to enrich the semantics of textless entities, thereby revealing deeper correlations among disparate facts [25,51]. However, the acquisition of relevant background knowledge remains under-explored in the TKGR task. To address this gap, we propose an entity-oriented adaptive knowledge construction strategy within the knowledge module, consisting of three components (i.e., *retrieval*, *generation*, and *refining*), to provide expressive background knowledge for each entity.

Among various knowledge graph databases, CONCEPTNET [52] is recognized as a widely used commonsense knowledge graph. It is constructed from free-form texts and stores plenty of structured commonsense information represented as words or phrases. Therefore, we employ the CONCEPTNET as the external knowledge base (KB), and concurrently develop an entity-oriented knowledge construction strategy to acquire the associated background knowledge for each entity. Concretely, given a query $q = (s, r, o, t)$, we use the objective entity o as an example to illustrate the construction process for obtaining the background knowledge.

Retrieval. We first convert the entity o into a concept, referred to as a zero-hop concept, and then match the entity text with the concept tokens in CONCEPTNET. Additionally, we incorporate heuristic rules, such as soft matching with lemmatization and filtering of stop words, to enhance the matching process. In this work, our focus lies in extracting one-hop concepts and all relations associated with the zero-hop concept. Consequently, the structural background knowledge of the entity o consists of the zero-hop concept, one-hop concepts, and all relations between zero-hop concept and one-hop concepts.

Generation. If the background knowledge cannot be retrieved from the KB, we resort to the generation method, which involves generating one-hop concepts using the given relation existing in CONCEPTNET. Specifically, we primarily utilize a pre-trained model, COMET [53], which is specifically designed for the knowledge base completion task. COMET is constructed by fine-tuning GPT [54] on CONCEPTNET. In this process, we use the zero-hop concept o in conjunction with a candidate relation $r_i^c \in \mathcal{R}^c$ as input to COMET, yielding a generated one-hop concept e. Here, $\mathcal{R}^c$ denotes the set of candidate relations maintained in

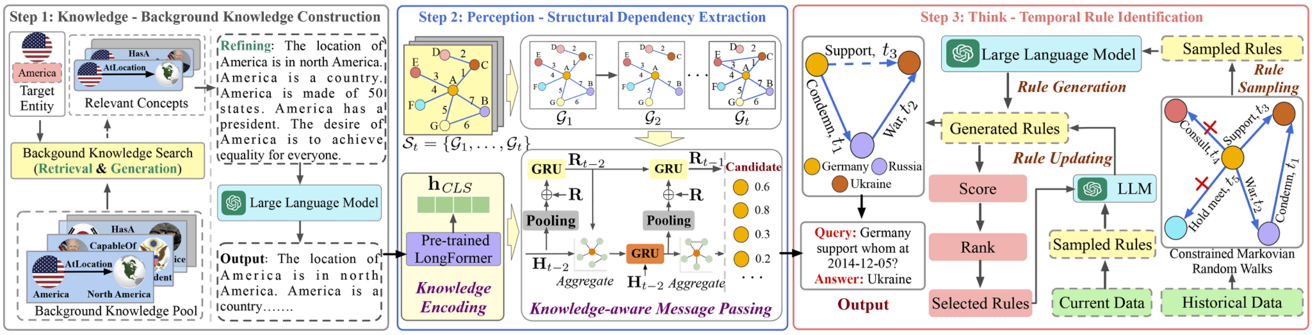

Fig. 1. Illustration of the proposed EV-COT model. (1) Detailed process of background knowledge construction in the knowledge module, which generates semantic-rich contextual information for each entity. (2) Detailed process of structural dependency extraction in the perception module, which captures structural and temporal patterns from the TKG topology. (3) Detailed description of temporal rule identification in the think module, which constructs interpretable reasoning chains to support temporal inference.

CONCEPTNET. The specific candidate relations adopted in this work are detailed in the experimental settings (see Section 5.1.4). By leveraging COMET, we obtain a complete set of concepts and relations as the background knowledge for the given entity o. Furthermore, let us provide an example to illustrate the acquired structural background knowledge. For instance, when considering the entity “François Hollande”, the following knowledge can be obtained: *(François Hollande, CapableOf, Be President)* and *(François Hollande, LocatedNear, Paris)*.

Refining. Upon obtaining structural knowledge (s_k, r_k, o_k) from CONCEPTNET, where s_k corresponds to the head entity, similar to s in a query $q = (s, r, o, t)$, r_k represents the relation, and o_k denotes the tail entity, we manually construct an associated relation template for each relation $r_k \in \mathcal{R}^c$ to express the comprehensive semantics of the knowledge triple. For instance, the triples *(François Hollande, CapableOf, Be President)* and *(François Hollande, LocatedNear, Paris)* can be expressed as “François Hollande is capable of being president” and “François Hollande is located near Paris”, respectively. These templates are then combined into a text sequence, forming knowledge prompts $\mathcal{P}$. However, the generated knowledge prompts frequently contain chaotic and misleading content, making reasoning challenging, thus requiring the reorganization of the prompts. To address this issue, we empower the LLM $\mathcal{F}(\cdot)$ (i.e., ChatGPT [48]) focus on knowledge elements while restore and filter the original prompt content, thereby generating refined textual content $\mathcal{P}_r$. This step can be formulated as follows:

$$\mathcal{P}_r = \mathcal{F}\left(\mathcal{P}; \mathrm{Prompt}_1\right). \tag{2}$$

The refined knowledge prompt $\mathcal{P}_r$ is subsequently utilized as input to the next stage, aiming to enhance the modeling of structural dependencies for the TKGR task. Simple example of the refined knowledge prompt Prompt_1 is provided in the following prompt box:

Prompt for the refining in the knowledge stage.

Task Instruction: You are an expert in the temporal knowledge graph reasoning, and please analyze these generated background knowledge identify and revise any low-quality content, correct errors, and enhance coherence while preserving the essential details. Input: {. . . .}

4.2. Perception - structural dependency extraction

In this module, the generated background knowledge is integrated into the structural dependency learning process, facilitating more comprehensive topology extraction through knowledge-aware message passing.

Knowledge Encoding. To extract valuable semantic information from the background knowledge, we leverage widely used pre-trained language models (PLMs), such as Longformer [55], to encode the background knowledge and obtain expressive semantic representations for each entity. Given an entity o and its corresponding knowledge prompts $\mathcal{P}_r^o$, we employ the knowledge prompt $\mathcal{P}_r^o$ to replace the entity o. Following the strategy in PLMs [55], we add a special symbol [CLS] at the beginning of the knowledge prompt $\mathcal{P}_r^o$. Consequently, the knowledge prompt can be transformed into $\mathcal{K}_o = \{[\mathrm{CLS}], w_1, \ldots, w_m\}$, where m denotes the token length of $\mathcal{P}_r^o$. Afterwards, we feed the concatenated knowledge prompt $\mathcal{K}_o$ into the Longformer model, and obtain the knowledge-aware semantic representations for the corresponding entity o:

$$[\mathbf{h}_{CLS}, \mathbf{h}_1, \ldots, \mathbf{h}_m] = \mathrm{Longformer}([\mathrm{CLS}], w_1, \ldots, w_m), \tag{3}$$

where $\mathbf{h}_i \in \mathbb{R}^d$ denotes the i-th token's representation. Following PLMs [55], we utilize the special token [CLS] representation $\mathbf{h}_{CLS}$ as the entity o's representation $\boldsymbol{v}_o$.

Knowledge-aware Message Passing. Structural dependencies capture the interaction correlations among co-occurring events, including the associations between entities through factual interactions and the semantic correlations among relations via shared entities. These dependencies play a crucial role in the TKGR task [8,11]. To enhance the modeling of these dependencies, we further incorporate background knowledge into the structural dependencies learning process, enabling more comprehensive and knowledge-aware interaction correlations extraction. Concretely, for a given entity o, we initialize its representation using its entity-oriented knowledge embedding $\boldsymbol{v}_o$. We then employ a relation-aware graph convolutional network (GCN), i.e., COMPGCN [56], to integrate relational information into the entity-level message aggregation process. Given a knowledge graph snapshot G_t, the representation of the entity o is iteratively updated by aggregating messages from neighboring subject entities s and their corresponding relations r via a message-passing mechanism at the l-th layer, where $l \in [0, L]$. Formally, this aggregation process in the GCN can be expressed as follows:

$$\mathbf{h}_{o,t}^{l+1} = \sigma\left(\frac{1}{c_o} \sum_{(s,r,o,t) \in G_t} \mathbf{W}_r^l\left(\vartheta\left(\mathbf{h}_{s,t}^l, \mathbf{r}_t\right)\right) + \mathbf{W}_o^l \mathbf{h}_{o,t}^l\right), \tag{4}$$

where $\mathbf{h}_{s,t}^l$ and $\mathbf{h}_{o,t}^l$ denote the l-th layer embeddings of entities s and o, respectively. $\mathbf{r}_t$ represents the relation embedding. $\mathbf{W}_r^l$ and $\mathbf{W}_o^l$ represent the l-th layer learnable parameters. The normalization constant c_o is equal to the in-degree of entity o. σ is the Relu activation function, and ϑ denotes the one-dimensional convolution operator, which constructs the translational property between subject entity s and relation r.

Sequential temporal patterns can reflect diverse behavioral trends and preferences underlying factual dynamics. To effectively extract informative historical cues from previous facts, we design a double-gated recurrent strategy, employing GRUs to progressively model sequential dependencies from both entity and relation perspectives-namely, an entity-wise GRU and a relation-wise GRU. Specifically, the entity-wise GRU is utilized to update the embeddings of entities across the sequence of subgraphs, as formulated below:

$$\mathbf{H}_t = \mathrm{GRU}\big(\mathbf{H}_{t-1}, \mathbf{H}_{t-1}^{GCN}\big), \tag{5}$$

where $\mathbf{H}_t$ and $\mathbf{H}_{t-1} \in \mathbb{R}^{|\mathcal{R}|\times d}$ represent the entity embedding matrices at the timestamps t and $t-1$, respectively. $\mathbf{H}_{t-1}^{GCN}$ denotes the entity embedding matrix learned by the GCN at timestamp $t-1$. To ensure consistency with the update of entity embeddings in the subgraph sequence, the relation-wise GRU is used to update relation representations:

$$\begin{aligned} \mathbf{r}'_t &= \big[\text{pooling}\left(\mathcal{H}_{r,t}\right); \mathbf{r}\big], \\ \mathbf{R}_t &= \mathrm{GRU}\left(\mathbf{R}_{t-1}, \mathbf{R}'_t\right), \end{aligned} \tag{6}$$

where $\mathcal{H}_{r,t}$ represents the embedding matrix of all entities connected to relation r at the timestamp t, extracted from $\mathbf{H}_t$, and $\mathbf{r}'_t$ is obtained by concatenating the mean pooling results of $\mathcal{H}_{r,t}$ and $\mathbf{r}$. $\mathbf{R}'_t$ consists of r'_t for all relations, while $\mathbf{R}_t$ and $\mathbf{R}_{t-1}$ are relation embedding matrices at timestamps t and $t-1$, respectively. Finally, the relation embedding $\mathbf{R}_t$ is updated using $\mathbf{R}_{t-1}$ and $\mathbf{R}'_t$ through a relation-wise GRU.

4.3. Thinking - temporal rule identification

Temporal logical rules are instrumental in uncovering temporal patterns and promoting interpretable reasoning for the TKGR task [38].

Rule Sampling. To exploit the logical correlations among historical facts, we propose a constrained Markovian random walk that simultaneously accounts for two critical factors (i.e., temporal order and temporal intervals) when selecting the subsequent fact during the traversal process. Specifically, we first perform random walks under Markovian assumptions to identify closed temporal paths and subsequently filter the resulting paths based on temporal constraints. Given an edge $r(s,o,t_l)$ as the target rule, we employ Markovian random walks to iteratively sample edges from the TKG $\mathcal{G}$ that are adjacent to the current object entity, continuing the traversal until a walk of length $l+1$ is obtained. For sampling step $m \in \{2,\dots,l+1\}$, let $\tilde{r}(s_e,o_e,t)$ represent the previously sampled edge, and let $\mathcal{A}(m,o_e,t)$ be the set of feasible edges for the next transition. To enforce the Markovian constraints, $\mathcal{A}(m,o_e,t)$ can be defined as follows:

$$\mathcal{A}\big(m,o_e,t\big) := \begin{cases} \big\{r_m\big(o_e,e,\hat{t}\big) \mid r_m\big(o_e,e,\hat{t}\big) \in \mathcal{G}\big\} & \text{if } m=2, \\ \big\{r_m\big(o_e,e,\hat{t}\big) \mid r_m\big(o_e,e,\hat{t}\big) \in \tilde{\mathcal{G}}\big\} & \text{if } m \in \{3,\dots,l\}, \\ \big\{r_m\big(o_e,s,\hat{t}\big) \mid r_m\big(e_o,s,\hat{t}\big) \in \tilde{\mathcal{G}}\big\} & \text{if } m=l+1, \end{cases} \tag{7}$$

where $\tilde{\mathcal{G}} := \mathcal{G}\backslash\big\{\tilde{r}^{-1}\left(o_e,s_e,t\right)\big\}$ excludes the inverse edge to avoid generating redundant rules. To ensure that the walks are cyclic, in the final sampling step $m=l+1$, we require sampling an edge that reconnects the walk to the initial entity s, if such an edge exists. Otherwise, we sample the next walk. The temporal paths sampled by all Markovian random walks are aggregated to construct the candidate path set $\mathcal{M}_r$. Subsequently, we introduce a temporal filtering operator $\mathcal{I}(t)$ for $\mathcal{M}_r$, which leverages temporal order to refine the set of candidate paths.

$$\mathcal{I}(t) = \begin{cases} 1, & \text{if } t < t_l \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

After obtaining the filtered candidate paths, we impose temporal intervals as an additional constraint. This constraint guides the selection of the next node based on a transition probability $P_{so}(t)$, defined as:

$$P_{so}(t) = \frac{\exp(-\lambda(t-T))}{\sum_{\left(s,o,t'\right)\in\mathcal{G}} \exp\left(-\lambda(t'-T)\right)}, \tag{9}$$

where $P_{so}(t)$ represents the probability of transitioning from node s to node o at time t, t is the timestamp of the edge, and T is the current time. $\exp(-\lambda(t-T))$ denotes an exponential decay function, which assigns weights to edges based on their temporal distance from the current timestamp T. λ is the decay rate parameter that controls the emphasis placed on more recent events.

Rule Generation. Through constrained Markovian random walks over historical events, EV-COT effectively captures relevant temporal rules $\mathcal{S}$ from the sampled temporal paths. Next, we leverage the LLM with strong reasoning capabilities to capture crucial evidence and salient temporal dependencies from the sampled temporal paths $\mathcal{S}$, and then generate high-coverage, high-quality general rules $\mathcal{S}_g$. A simple example of the prompt is provided in the following prompt box:

Prompt for Rule Generation.

Task Instruction: You are an expert in TKGR, and please generate as many temporal logical rules as possible related to 'r' based on extracted temporal rules. For the relations in rule body, you are going to choose from the candidate relations. Input: {… .}

Rule Updating. Due to the evolving nature of TKGs, the constructed rules $\mathcal{S}_g$ tend to remain static and may gradually degrade into low-quality rules, rendering them less effective for reasoning over newly emerging events. To address this issue, we design an adaptive update strategy that generates rules from the most recent event data and employs them as references to refine existing low-quality rules. This strategy enables the extracted rules to continuously integrate new knowledge, thereby adaptively supporting future event prediction. Specifically, EV-COT first employs the *Confidence* metric [57] to measure the reliability of a temporal rule $\rho \in \mathcal{S}_g$. This metric is defined as the proportion of temporal fact pairs that satisfy the rule body *rule_body*(ρ) and also satisfy the entire *rule*(ρ):

$$c_\rho = \frac{\text{Number of fact pairs satisfying } \mathit{rule\_body}(\rho)}{\text{Number of fact pairs satisfying } \mathit{rule}(\rho)}, \tag{10}$$

where c_ρ denotes the confidence of rule ρ. A higher confidence value indicates greater reliability of the rule ρ. Next, we select a subset of rules with low confidence, defined as $\mathcal{S}_g^{\text{low}} = \{\rho \in \mathcal{S}_g \mid c_\rho < \gamma\}$, where γ is the confidence threshold for identifying low-quality rules. Subsequently, EV-COT employs Markovian random walks to extract temporal logical rules from the current data, which are then used to refine the low-confidence rules in $\mathcal{S}_g^{\text{low}}$. This process yields an updated rule set, denoted as $\mathcal{S}_d$, after $\mathcal{N}$ iterations.

4.4. Entity reasoning

Entity reasoning aims to infer the potential entity o for a given query $q=(s,r,?,T)$ by integrating the diverse reasoning results from both the perception and thinking modules. Accordingly, our EV-COT model is equipped with two key reasoning components: Rule-wise Reasoning and Graph-wise Reasoning (Algorithm 1).

Rule-wise reasoning primarily utilizes the generated high-score rules in the thinking module to conduct in-depth logical reasoning for future event prediction, deducing new entities as potential answers [39]. Given a query $(s,r,?,T)$, EV-COT scores the rule through the Eq. (10), and then select the high-scoring rules:

$$\mathcal{N}_s = \big\{\rho \mid c_\rho > \gamma, \rho \in \mathcal{S}_d\big\}, \tag{11}$$

where $\mathcal{N}_s$ denotes the high-confidence rule set, comprising the rules ρ whose confidence scores satisfy $c_\rho > \gamma$, with γ being the predefined threshold. According to the rule $\rho \in \mathcal{S}_d$, we are able to obtain the reasoning paths and further derive the entity o:

$$(s,r,o,T) \leftarrow \wedge_{i=1}^{t-1}\big(s_i,r_i,o_i,i\big), \tag{12}$$

Algorithm 1 Reasoning process of EV-COT framework.

```
Input: Historical TKG snapshots S_{T-1}; Query q = (s, r, ?, T); External KB
    (e.g., CONCEPTNET); LLM F (e.g., ChatGPT).
Output: Predicted entity o for query q.
for entity e ∈ {s, o} do
    Retrieve 1-hop concepts from KB via soft matching;
    if retrieval fails then
        Generate concepts using COMET with relation templates;
    end if
    Refine knowledge prompts P_r ← F(P; Prompt_1);
end for
for entity e ∈ {s, o} do
    Encode P_r via Longformer;
    Initialize entities embeddings;
    Update embeddings via Eq. (4);
    Capture temporal patterns via GRU (Eq. (5));
end for
Output graph-based scores: Score_(graph,o) ← ⟨H_T^{to}, o⟩;
Perform constrained Markovian walks to extract temporal paths;
Generate rules;
Update rules S_d via confidence filtering (Eq. (8)) and
    dummyTXdummy-( decay weights;
Compute rule-based scores Score_(ρ,o) via Eq. (11);
Compute Score ← β · Score_(ρ,o) + (1 − β) · Score_(graph,o);
Rank candidates and return top-k entities;
```

where o denotes the candidate derived based on the rule ρ. Due to the time decay property inherent in temporal data, we further select candidate entities that are most relevant to the given query:

$$\text{Score}_{(\rho,o)} = \sum_{\rho \in \mathcal{N}_s} \sum_{body(\rho)(s_t,o_t,t)\in\mathcal{G}} (c_\rho + \exp(-\lambda(t-T))), \tag{13}$$

where $\text{Score}_{(\rho,o)}$ denotes the score of candidate entity o, computed based on the searched path in the TKG using rule ρ at the corresponding timestamp T. c_ρ denotes the confidence of the temporal rule ρ, λ is the decay rate parameter, and $body(\rho)(s_t, o_t, t) \in \mathcal{G}$ indicates that the rule body is satisfied by a path in the TKG at timestamp t.

In addition, we introduce a graph-wise reasoning strategy that operates on the representations $\mathbf{H}_T$ generated by the perception module. This strategy enables the prediction of candidate answers for a given query, serving as a complementary mechanism when the extracted rules fail to fully cover all correct answers. The score can be computed through the inner product operation:

$$\text{Score}_{(graph,o)} = \langle \mathbf{H}_T, o \rangle. \tag{14}$$

Since the candidates of rule-based reasoning $\mathcal{E}_{(\rho,o)}$ and graph-based reasoning $\mathcal{E}_{(graph,o)}$ may partially overlap and differ, we assign a score of $\text{Score}_{(\rho,o)} = 0$ for entities $o \in \mathcal{E}_{(graph,o)}$ but $o \notin \mathcal{E}_{(\rho,o)}$, and vice versa. The overall score for each candidate entity is computed as follows:

$$\text{Score} = \beta \cdot \text{Score}_{(\rho,o)} + (1-\beta) \cdot \text{Score}_{(graph,o)}, \tag{15}$$

where Score represents the final score of the candidate o from rule-based reasoning and graph-based reasoning modules. β is used to assign weights to different scores. Finally, EV-COT aggregates all candidates, and then sorts these candidates to select those that best meet the query requirements.

4.5. Model analysis

4.5.1. Relation with COH & GPT-NeoX

From the perspective of COT, GPT-NeoX primarily relies on in-context learning to capture contextual knowledge by retrieving first-order past facts associated with the query. Notably, GPT-NeoX does not employ chain-of-thought reasoning. In addition, COH directly utilizes chain-of-thought reasoning to integrate first-order and second-order fact histories relevant to the query through step-by-step reasoning. Unlike COH, EV-COT employs an event-aware chain-of-thought mechanism to mimic the human thinking process, including acquiring background knowledge, understanding context, and performing logical reasoning, thereby enabling deep reasoning about event evolution through a three-stage hierarchical reasoning process.

From the perspective of information, GPT-NeoX and COH consider only first-order or second-order structural information in TKGs. Moreover, because only limited historical facts are fed into the language model, these baselines fail to capture the complex structural dependencies and temporal interactions inherent in TKGs. In contrast, EV-COT incorporates an event-aware chain-of-thought approach that not only models complex structural-temporal dependencies across TKGs but also integrates background knowledge and temporal logical rules to enhance TKG reasoning. As a result, EV-COT captures richer event cues that improve prediction accuracy, serving as an extended and more expressive version of GPT-NeoX and COH.

From the perspective of model architecture, GPT-NeoX and COH primarily transform graph structures into textual sequences and then directly apply LLMs for reasoning. In contrast, we combine LLMs with graph neural networks and disentangle the reasoning process into three stages: the knowledge module retrieves entity-level commonsense knowledge; the perception module, equipped with a relation-aware graph encoder, extracts graph-level structural-temporal patterns; and the thinking module, incorporating an adaptive update strategy, mines crucial temporal logical rules and iteratively refines them to enable deep reasoning about event evolution. Based on experimental results, EV-COT demonstrates stronger scalability compared to GPT-NeoX and COH on larger datasets in the TKGR task.

5. Experiments

In this section, we evaluate the effectiveness of EV-COT through comprehensive experiments conducted on three real-world datasets in the TKGR task. We perform a comparative experimental analysis, ablation study, and case study to discuss the importance of each component in EV-COT.

5.1. Experimental setup

5.1.1. Datasets

To analyze the effectiveness of EV-COT in the field of TKGR, we select three representative benchmark datasets: ICEWS14 [58], ICEWS18 [15], and ICEWS15 [58]. The statistical details of datasets are reported in Table 1, followed by a comprehensive description of each dataset below:

- **ICEWS18** refers to a subset of data derived from the Integrated Crisis Early Warning System (ICEWS) [59]. This dataset captures factual

Table 1
Statistics of the datasets. # represents the quantity.

Dataset	#Entities	#Relations	#Train	#Validation	#Test	#Timestamps	Time interval
ICEWS18	23,033	256	373,018	45,995	49,545	7,272	24 hours
ICEWS14	7,128	230	74,845	8,514	7371	365	24 hours
ICEWS15	10,488	251	368,868	46,302	46,159	4,017	24 hours

events that took place in 2018, with a daily granularity. It encompasses a total of 23,033 unique entities and 256 different types of relations.
- **ICEWS14** represents an alternative version of the ICEWS dataset, focusing on concurrent events occurred in 2014. This particular dataset encompasses a total of 7128 unique entities and captures 230 distinct types of relations among different entities.
- **ICEWS15** is an extensive dataset obtained through scraping the ICEWS database. It contains a wealth of factual information spanning from January 1, 2005, to December 31, 2015, with a daily granularity. This dataset comprises 10,488 unique entities and encompasses 251 different types of relations.

Furthermore, following existing work [15], we follow the consistent pre-processing strategy for all datasets. With the exception of ICEWS14, each dataset is divided into training, validation, and test sets, with a proportion of 80 %, 10 % and 10 % by timestamps. Therefore, this operation ensures that the temporal order is maintained across datasets, and we can observe: (time stamps of the train) < (time stamps of the valid) < (time stamps of the test).

5.1.2. Baseline

To evaluate the effectiveness of our EV-COT in entity prediction, we compare it against a set of 16 competitive knowledge graph reasoning models, comprising both static and temporal approaches. Specifically, the static KG reasoning methods include ComplEx [60], R-GCN [37] and ConvE [61]. For temporal KG reasoning methods with the interpolation setting, we select HyTE [28], TTransE [62] and TA-DistMult [63]. Furthermore, we utilize existing TKGR methods with the extrapolation setting, involving RE-NET [15], RE-GCN [11], CyGNet [36], xERTE [38], TANGO [40], TiRGN [8] and CENET [9]. LLM-based methods for the TKGR task include PPT [47], GPT-NeoX [10], CoH [21], and MESH [21].

- **ComplEx** [60] presents a simple matrix and tensor factorization for prediction. It utilizes vectors with complex values and maintains the mathematical definition of the dot product.
- **R-GCN** [37] introduces relational graph convolutional networks designed for effectively handling multi-relational data in the field of knowledge graph reasoning.
- **ConvE** [61] designs a multi-layer 2D convolutional network to effectively capture entity and relation representations within knowledge graphs.
- **HyTE** [28] introduces a temporal knowledge graph embedding framework based on hyperplanes. It aims to jointly encode entity/relation and temporal information as hyperplane vectors across different TKG snapshots.
- **TTransE** [62] is an extension of the popular embedding model TransE [64] to address the temporal knowledge graph reasoning by replacing its scoring function.
- **TA-DistMult** [63] mainly utilizes recurrent neural networks to capture time-varying entity and relation representations incorporating temporal information for reasoning in temporal knowledge graphs.
- **RE-NET** [15] incorporates a graph neural network-based neighborhood aggregator to capture event connections within the same time window for TKGR.
- **RE-GCN** [11] employs a relation-aware graph convolutional network (GCN) to capture the structural dependencies, i.e., the evolutional representations of entities and relations, within a sequence of events from knowledge graphs.
- **CyGNet** [36] introduces a copy-generation mechanism composed of two modules: a copy module and a generation module, which together predict entity probabilities from a known entity vocabulary.
- **xERTE** [38] generates query subgraphs with certain hop numbers by constructing inference graphs and designs a temporal relational attention mechanism to guide the extraction of an enclosing subgraph around the query.
- **TANGO** [40] expands the concept of ordinary differential equations to multi-graph convolutional networks, enabling the preservation of continuity in dynamic multi-graph and facilitating the generation of continuous-time embeddings.
- **TiRGN** [8] designs a local-global temporal knowledge graph encoder framework for the TKGR task. It utilizes a local recurrent graph network to capture the historical dependency of events and meanwhile uses the global history encoder network to collect repeated historical facts.
- **CENET** [9] utilizes contrastive learning to capture both historical and non-historical dependencies. It then trains a classifier that outputs a boolean value to determine the entities that require greater attention.
- **PPT** [47] introduces pre-trained language models (PLMs) for future event prediction. It designs various time-specific prompts for different types of timestamps and trains the PLMs using a masking strategy.
- **GPT-NeoX** [10] reformulates TKG prediction as an in-context learning (ICL) task by providing LLMs with the first-order history of the query in textual form, enabling them to generate plausible answers without any fine-tuning.
- **CoH** [21] designs a Chain-of-History reasoning mechanism that explores important high-order historical chains in a step-by-step manner and infers answers to queries based solely on the history chains derived in the final step. Notably, LLaMA-2-7B-CoH, Vicuna-7B-CoH, and Mixtral-CoH are variant models built upon the CoH framework.
- **MESH** [21] employs two types of expert modules (i.e., GNN and LLM) to integrate both structural and semantic information, thereby guiding the reasoning process for diverse events.

5.1.3. Evaluation metrics

Following existing works [8,11,15], we evaluate the performance of our model on the entity prediction task using two commonly used metrics: mean reciprocal rank (MRR) and Hits@1/3/10 (H@1/3/10). Following the prior works [11], we utilize the time-aware filtered setting to report the experimental results.

5.1.4. Model implementation

In our EV-COT smodel, we configure several key hyperparameters for different components. Specifically, EV-COT sets the decay rate λ during the logical rule sampling process, the threshold θ for dynamic adaptation, and the minimum confidence γ for candidate generation. The values are set as follows for both datasets: $\lambda = 0.1$, $\theta = 0.01$, and $\gamma = 0.01$. The number of iterations for the dynamic adaptation process is set to 5. For the background knowledge, we selected 20 candidate semantic relations from CONCEPTNET that are deemed potentially beneficial for TKGR: *AtLocation, CapableOf, Causes, CausesDesire, DesireOf, Desires, HasA, HasProperty, InheritsFrom, IsA, LocatedNear, LocationOfAction, MotivatedByGoal, NotHasA, NotHasProperty, NotIsA, PartOf, ReceivesAction, RelatedTo, SymbolOf*. For the perception module, we utilize the pre-trained TiRGN model. For the thinking module, we leverage a local large language model (Qwen2.5-7b[1], Qwen2.5-3B[2] and LLama3.1-8B[3]). For baseline comparisons, we strictly adhere to the parameter settings outlined in their respective original papers. All experiments are conducted on a system comprising an Intel(R) Core(TM) i7-13700KF processor running at 3.40 GHz, equipped with two NVIDIA GeForce RTX 4090 GPUs with 24 GB of VRAM, and accompanied by 128 GB of system RAM.

[1] https://huggingface.co/Qwen/Qwen2.5-7B
[2] https://huggingface.co/Qwen/Qwen2.5-3B
[3] https://huggingface.co/meta-llama/Llama-3.1-8B

Table 2
Performance (in percentage) comparison on ICEWS18, ICEWS14, and ICEWS15 with time-aware filtered setting. The best results are in **bold** font and the seconds are underlined. Higher values of MRR, H@1, H@3 and H@10 indicate better performance.

Method	Train	ICEWS18				ICEWS14				ICEWS15			
		MRR	Hits@1	Hits@3	Hits@10	MRR	Hits@1	Hits@3	Hits@10	MRR	Hits@1	Hits@3	Hits@10
ComplEx	✔	15.45	8.04	17.19	30.73	22.61	9.88	28.93	47.57	20.26	6.66	26.43	47.31
R-GCN	✔	15.05	8.13	16.49	29.00	28.03	19.42	31.95	44.83	27.13	18.83	30.41	43.16
ConvE	✔	22.81	13.63	25.83	41.43	30.30	21.30	34.42	47.89	31.40	21.56	35.70	50.96
HyTE	✔	7.41	3.10	7.33	16.01	16.78	2.13	24.84	43.94	16.05	6.53	20.20	34.72
TTransE	✔	8.44	1.85	8.95	22.38	12.86	3.14	15.72	33.65	16.53	5.51	20.77	39.26
TA-DistMult	✔	16.42	8.60	18.13	32.51	26.22	16.83	29.72	45.23	27.51	17.57	31.46	47.32
RE-Net	✔	29.78	19.73	32.55	48.46	39.86	30.11	44.02	58.21	43.67	33.55	48.83	62.72
RE-GCN	✔	32.62	22.39	36.79	52.68	42.00	31.63	47.20	61.65	48.03	37.33	53.90	68.51
CyGNet	✔	27.12	17.21	30.97	46.85	37.65	27.43	42.63	57.90	40.42	29.44	46.06	61.60
xERTE	✔	29.31	21.03	33.51	46.48	40.79	32.70	45.67	57.30	46.62	37.84	52.31	63.92
TANGO	✔	28.97	19.51	32.61	47.51	36.94	27.60	41.10	54.41	42.86	32.72	48.14	62.34
TiRGN	✔	33.66	23.19	37.99	54.22	44.04	33.83	48.95	63.84	50.04	39.25	56.13	70.71
CENET	✔	31.43	23.47	34.05	47.27	38.58	30.18	41.79	55.20	46.34	38.14	49.89	62.53
PPT	✔	26.61	16.92	30.64	45.44	38.42	28.92	42.52	57.03	38.92	28.64	43.43	58.61
GPT-NeoX	✘	–	19.22	31.33	41.44	–	33.45	46.01	56.54	–	35.29	52.35	70.35
COH-Llama	✘	–	22.32	36.35	52.24	–	34.93	47.04	59.19	–	38.65	54.14	69.91
COH-Vicuna	✘	–	20.91	34.72	53.65	–	32.82	45.73	65.64	–	39.22	54.63	70.72
COH-Mixtral	✘	32.98	21.83	37.79	54.92	43.94	33.07	49.64	64.90	49.71	38.01	56.40	71.25
MESH	✔	33.96	–	38.37	54.12	44.36	–	49.81	64.21	48.66	–	54.26	68.57
EV-COT (Qwen-7B)	✘	**35.84**	**24.06**	**41.73**	**56.95**	**47.12**	**35.97**	**52.34**	**68.56**	**53.54**	**41.23**	**58.43**	**75.78**
EV-COT (Qwen-3B)	✘	35.60	23.47	41.31	56.18	46.79	35.11	51.79	67.61	53.18	40.24	57.81	74.72
EV-COT (LLama-8B)	✘	35.77	23.70	41.61	56.60	47.03	35.43	52.19	68.14	53.44	40.58	58.27	75.32

Table 3
Ablation study of EV-COT.

Model	ICEWS18				ICEWS14				ICEWS15			
	MRR	Hits@1	Hits@3	Hits@10	MRR	Hits@1	H@3	Hits@10	MRR	Hits@1	Hits@3	Hits@10
EV-COT	**35.84**	**24.06**	**41.73**	**56.95**	**47.12**	**35.97**	**52.34**	**68.56**	**53.54**	**41.23**	**58.43**	**75.78**
w/o knowledge	34.75	23.82	39.34	55.25	46.16	35.19	50.72	66.85	52.52	39.90	57.46	75.03
w/o perception	34.18	23.56	38.45	54.70	44.49	34.44	49.60	64.52	50.15	39.23	56.29	71.52
w/o thinking	34.07	23.47	38.32	54.59	44.43	34.33	49.50	64.36	50.03	39.12	56.08	71.27
w historical	34.27	22.79	39.72	55.39	44.84	33.52	49.69	66.62	51.96	39.30	56.20	73.90
w current	34.64	22.85	40.16	55.53	45.17	34.06	50.25	66.97	52.30	39.81	56.91	74.73
w H+C	34.88	23.08	40.49	55.89	45.45	34.27	50.65	67.07	52.68	39.89	57.27	75.11
BERT	35.75	24.01	41.62	56.84	47.01	35.90	52.21	68.43	53.41	41.15	58.28	75.64
GPT-2	36.14	23.30	42.92	57.59	47.03	35.88	52.27	68.45	53.43	41.12	58.36	75.65

5.2. Performance comparison

The comprehensive performance of EV-COT, along with several state-of-the-art models on three benchmark datasets, is presented in Table 2. Based on these experimental results, the following observations have been derived:

(O1) According to the performance comparison, it is evident that EV-COT exhibits substantial improvements compared to all baselines across all datasets. Notably, compared to the second-best model, EV-COT achieves an average relative improvements of 6.81 %, 3.50 %, 6.28 %, and 4.82 % in terms of MRR and Hit@K (i.e., Hit@1, Hit@3, and Hit@10). These results affirm that our EV-COT model is capable of capturing crucial evidence of event evolution through the event-aware chain-of-thought strategy, even without any additional training process. Compared to all strong baselines, our model generates more accurate reasoning results for the TKGR task, attributed to its step-by-step reasoning chains. Specifically, the knowledge module constructs high-quality background knowledge to enhance the semantic representation of each entity; the perception module captures structural and temporal patterns from the perspective of graph topology; and the thinking module adaptively constructs temporal logical rules to enable deep and interpretable reasoning.

(O2) For static KG models that consider only structural information, their performance is significantly inferior to that of their temporal counterparts, i.e., temporal KG reasoning methods. This discrepancy arises because static models capture solely structural attributes from static knowledge graph snapshots while failing to account for temporal dependencies among facts across different timestamps, ultimately resulting in suboptimal reasoning capabilities. These results further validate the effectiveness and rationality of our perception module design, which integrates both structural information and temporal dependencies to enhance predictive performance.

(O3) Notably, LLM-based methods generally outperform strong non-LLM baselines, highlighting the effectiveness of large language models in complex reasoning tasks such as TKGR. This observation further substantiates our design choice in the thinking module to introduce the LLM for iteratively capturing and updating temporal rules to enable interpretable reasoning. Meanwhile, our proposed method, EV-COT, achieves significantly better performance than existing LLM-based methods. We attribute this performance gain to the ability of EV-COT to

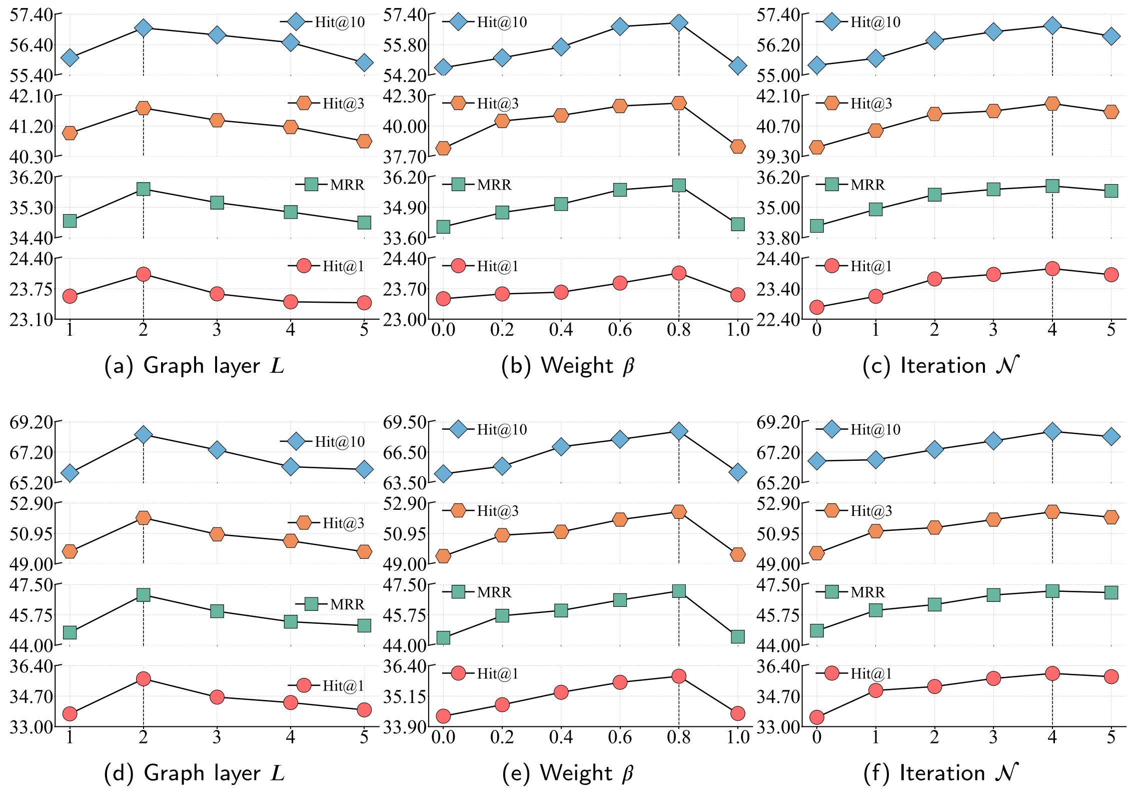

Fig. 2. Results of hyper-parameters L, β and $\mathcal{N}$ of EV-COT on ICEWS18 and ICEWS14. The first line (a-c) represents the ICEWS18's experimental results. The second line (d-f) denotes the analysis results on ICEWS14.

overcome the key limitations inherent in prior approaches: the inherent discrepancy between structured graph data and natural language.

5.3. Ablation study

To comprehensively assess the efficacy of each key component in EV-COT, we performed an ablation study using the ICEWS18, ICEWS14 and ICEWS15 datasets. The experimental results are presented in Table 3.

5.3.1. Impact of the knowledge module

To evaluate the contribution of background knowledge to the overall performance of EV-COT, we develop a variant model without the knowledge module (*w/o knowledge*). As shown in Table 3, removing the background knowledge leads to a decline in performance. This result suggests that extracting rich semantic information from external knowledge sources is beneficial for enhancing the effectiveness of EV-COT.

5.3.2. Impact of the perception module

To evaluate the impact of the perception module on TKGR, we develop a variant model by excluding the perception module (*w/o perception*). The substantial performance drop observed after its removal highlights the critical role of modeling interactive relationships between facts for effective reasoning in TKGs. Conversely, incorporating the perception module significantly boosts the model's performance. This result indicates that the perception module effectively captures fine-grained structural and temporal attributes, functioning as a form of data augmentation within EV-COT.

5.3.3. Impact of the thinking module

To demonstrate the contribution of the thinking module to the overall performance of the model, we construct a variant without the thinking module (*w/o thinking*). As shown in Table 3, the removal of the thinking module results in substantial performance degradation, underscoring the importance of modeling temporal logical rules for the TKGR task.

Table 4
Visualization experiment for the query *(Malaysia, Yield, ?, 2014-12-14)* with and without the perception module. The logic rules are generated by the thinking module. "America" is the actual answer.

Query	(Malaysia, Yield, ?, 2014-12-14)
Generated Rules	1. Mediate & Appeal to yield & Accede to demands for rights 2. Engage in negotiation & Consult 3. Engage in diplomatic cooperation & Demand meeting, negotiation 4. Demonstrate for leadership change & Make statement & Consult & Make a visit 5. Express accord
w/o perception	1. Philippines 2. **America** 3. Indonesia 4. Thailand 5. Vietnam
w perception	1. **America** 2. Indonesia 3. Thailand 4. Laos 5. Vietnam

5.3.4. Impact of the adaptive update strategy in the thinking module

The adaptive update strategy leverages current data to refine the rules initially generated by LLMs from historical data. To assess its effectiveness, we design three variants: using only historical data (*w historical*), using only current data (*w current*), and using both historical and current data (*w* $H+C$). Based on the experimental results, the variant achieves, w.r.t, *w* $H+C$, the best performance, suggesting that large-scale historical data provides generalizable background knowledge, while current data contributes contextually relevant information.

Table 5
Visualization experiment for the query (Japan, Make pessimistic comment, ?, 2014-12-09). Rules for rule generation and rules for adaptive update represent the dynamic changes of the rules during the rule generation and adaptive update. Candidates for rule generation and candidates for adaptive update represent the changes in the corresponding candidate rankings. “Australia” is the actual answer.

Query	(Japan, Make pessimistic comment, ?, 2014-12-09)
Rules for Rule Generation	1. Alleged violations of culture & Criticize or denounce 2. Use Unconventional Violence 3. Make an appeal or request & Consult 4. Make statement & Criticize or denounce 5. Meet at a ‘third’ location
Candidates for Rule Generation	1. Vietnam 2. India 3. **Australia** 4. Indonesia 5. Laos
Query	(Japan, Make pessimistic comment, ?, 2014-12-09)
Rules for Adaptive Update	1. Appeal for release of property & Make an appeal or request & Consult 2. Make statement & Criticize or denounce 3. inv Make statement & Express intent to meet or negotiate & Make A visit 4. Meet at a ‘third’ location 5. Alleged violations of culture & Criticize or denounce
Candidates for Adaptive Update	1. **Australia** 2. Indonesia 3. Philippines 4. Cambodia 5. India

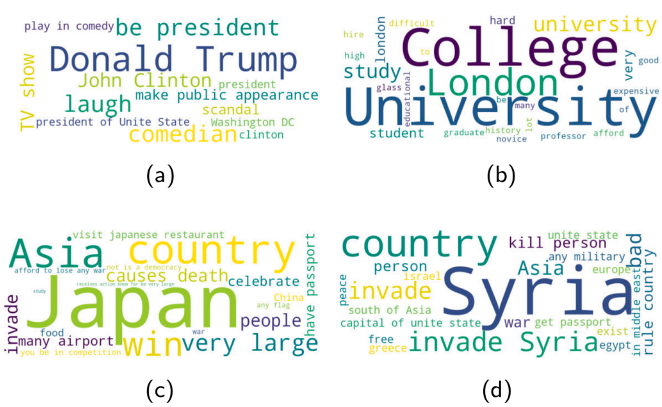

Fig. 3. Visualization of four instances for entity’s background knowledge using the word cloud. The first line (a-b) denotes the ICEWS14 and the second line (c-d) is from the ICEWS15 dataset. The largest and center word represents the selected target entity. The remaining words denote the corresponding background knowledge.

The combination of both enables more accurate and robust predictions of future events.

5.3.5. Impact of the knowledge encoding in the perception module

To evaluate the effectiveness of the knowledge encoder, we design three variant models by replacing Longformer with other pre-trained language models (i.e., BERT [44], GPT-2[4]. Experimental results show that all models exhibit comparable performance, with GPT-2 yielding a slight improvement on the ICEWS18 dataset. These findings indicate that the quality of the knowledge encoding has a relatively limited impact on the reasoning process in TKGR. Overall, Longformer achieves superior results on most datasets, confirming its suitability as the knowledge encoder for the TKGR task.

[4] https://huggingface.co/openai-community/gpt2

5.4. Hyper-parameter analysis

A sensitivity analysis of our EV-COT model is conducted to investigate the influence of four crucial hyperparameters: the number of graph layer L, the weight β on the losses and the iteration $\mathcal{N}$ in the thinking module on the ICEWS18 and ICEWS14 datasets. The sensitivity analysis results are reported in Fig. 2. Detailed findings are elucidated below.

• *The number of graph layer L.* For the hyper-parameter L, we vary the number of layers from 1 to 5. The results reveal that the performance of EV-COT initially improves with an increase in the number of layers L and subsequently declines when the layer is too large. Thus, we set the number of layers L to 2.

• *Weight β for the learning loss.* We vary the values of β in the set $\{0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$ to observe how the performance of EV-COT changes on the ICEWS18 and ICEWS14 datasets. The parameter β aims to balance the rule-based reasoning and graph-based reasoning. As depicted in Fig. 2 (b) and (e), we observe that the optimal value for β is 0.8.

• *Iteration $\mathcal{N}$ of the thinking module.* To investigate the impact of the number of iterations $\mathcal{N}$ in the thinking module, we explore values in the set $\{0, 1, 2, 3, 4, 5\}$. As shown in Fig. 2 (c) and (f), the optimal performance of EV-COT is achieved when the number of iterations in the thinking module is set to 4.

5.5. Case study

5.5.1. Analysis of background knowledge.

We randomly sample 4 data instances from ICEWS14 and ICEWS15 and visualize their corresponding background knowledge via the word cloud. As shown in Fig. 3, we can see clear correlations between the target entity and retrieved background knowledge. Therefore, the background knowledge is able to provide rich auxiliary information, which may not emerge in all datasets, for assisting model reasoning. This observation verifies the rationality and effectiveness of knowledge-enhanced TKG reasoning through the knowledge module.

5.5.2. Visualization of the changes in the perception module

To analyze the effectiveness of the perception module, we visualize experimental results obtained without and with the perception module.

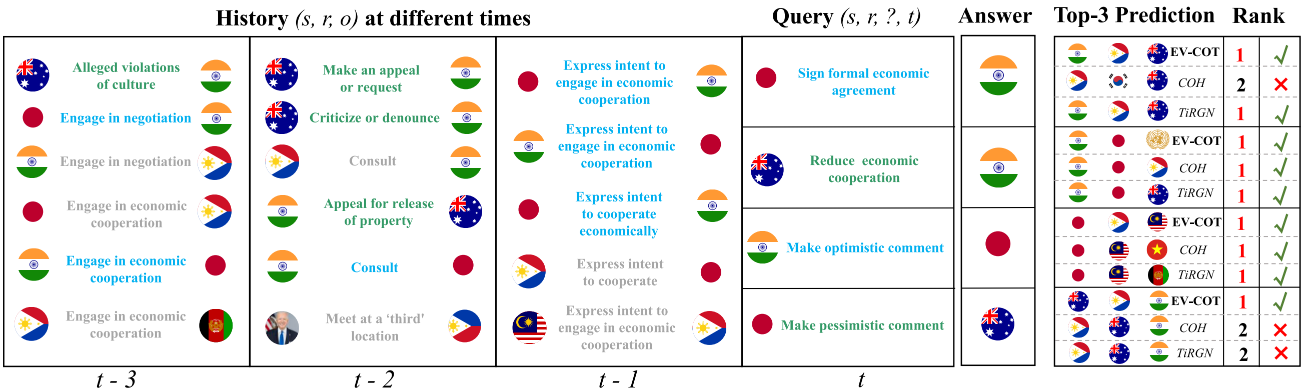

Fig. 4. Visualization of the model reasoning performance on the ICEWS14 dataset. we select four events related to the topic of "India" for analysis.

The results are summarized in Table 4. We observe that incorporating the perception module further improves accuracy on the TKGR task. This finding validates the effectiveness of our design in addressing the gap between graph structures and the language space by capturing structural-temporal dependencies within the perception module.

5.5.3. Visualization of the changes in the thinking module

To illustrate the impact of the adaptive update strategy within the *thinking* module, we present a case study in Table 5, highlighting changes in both rule and candidate rankings. For the query *(Japan, Make pessimistic comment, ?, 2014-12-09)*, the ground-truth answer *Australia* is initially ranked third during the rule generation phase. However, after applying the adaptive update process, it rises to the top-ranked position. In addition, the rankings of the generated rules exhibit notable reordering between the two stages. These observations demonstrate the effectiveness of the adaptive update strategy in refining LLM-generated reasoning paths by incorporating recent temporal context, thereby enhancing the model's ability to capture the dynamic and evolving nature of temporal knowledge graphs.

5.5.4. Analysis of model prediction performance

To analyze the prediction performances of our model and two best-performed baselines (i.e., TiRGN and COH), we randomly select four events related to the topic of "India" from the ICEWS14 test set. The results and prediction errors are presented in Fig. 4. Specifically, we report the Top-3 prediction results and visualize the discrepancies between our model and the baselines. The same colors across different events exhibit similar semantic correlations. Among the four future events in the Top-1 predictions, our model correctly predicts all of them, while TiRGN has one prediction error, and COH has two prediction errors. It is evident that TiRGN and COH lack the capability to identify relationships between entities. In contrast, our EV-COT demonstrates superior prediction results, highlighting the benefits of incorporating plug-and-play reasoning stage, i.e., knowledge, perception, and thinking.

6. Conclusion

In this work, we propose EV-COT, a novel event-aware chain-of-thought reasoning framework designed to enhance event evolution modeling. EV-COT consists of three plug-and-play components: a knowledge module, a perception module, and a thinking module. The knowledge module generates informative background knowledge to enrich the semantic representation of each entity; the perception module extracts structural and temporal cues from the perspective of graph topology; and the thinking module constructs temporal logical rules to enable interpretable reasoning. These modules collaboratively extract crucial event cues in a step-by-step reasoning manner to support effective inference. As future work, we plan to extend EV-COT to other domains, such as knowledge graph completion and KG-based recommender systems.

CRediT authorship contribution statement

Zhangtao Cheng: Writing – review & editing, Supervision, Investigation; **Shichong Li:** Visualization, Resources, Formal analysis; **Yichen Xin:** Writing – original draft, Methodology; **Bin Chen:** Validation, Methodology, Investigation; **Ting Zhong:** Supervision, Funding acquisition, Formal analysis; **Fan Zhou:** Supervision, Investigation, Funding acquisition, Formal analysis.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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